

📌 Bean Life Cycle Example in Spring (From Scratch)

- In this example, we will create a simple Maven Project and understand the Bean Life Cycle using:

✓ Constructor (Bean Creation)

✓ initMethod (Initialization)

✓ destroyMethod (Destruction)

- This is one of the easiest examples to demonstrate the Spring Bean Life Cycle in STS.

Project Structure

```
BeanLifeCycleDemo
|
├── src/main/java
|   |
|   └── com.vijaystack
|       |
|       ├── AppConfig.java
|       ├── EmployeeService.java
|       └── Test.java
|
└── pom.xml
```

Step 1: pom.xml

- Add Spring Context dependency.

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.vijaystack</groupId>
  <artifactId>BeanLifeCycleDemo</artifactId>
  <version>1.0</version>

  <dependencies>
```

```
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-context</artifactId>
  <version>6.2.8</version>
</dependency>

</dependencies>

</project>
```

Step 2: Create Bean Class

```
package com.vijaystack;

public class EmployeeService {

    public EmployeeService() {
        System.out.println("1. Bean Object Created");
    }

    public void init() {
        System.out.println("2. Initialization Method Called");
    }

    public void getEmployee() {
        System.out.println("3. Bean Ready For Use");
    }

    public void destroy() {
        System.out.println("4. Destruction Method Called");
    }
}
```

Step 3: Configuration Class

```
package com.vijaystack;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;

@Configuration
public class AppConfig {
```

```

@Bean(initMethod = "init", destroyMethod = "destroy")
public EmployeeService employeeService() {

    return new EmployeeService();
}
}

```

- ♦ initMethod executes after bean creation.
- ♦ destroyMethod executes when the container is closed.

Step 4: Test Class

```

package com.vijaystack;

import
org.springframework.context.annotation.AnnotationConfigApplicationContext;

public class Test {

    public static void main(String[] args) {

        AnnotationConfigApplicationContext context =
            new AnnotationConfigApplicationContext(AppConfig.class);

        EmployeeService service =
            context.getBean(EmployeeService.class);

        service.getEmployee();

        context.close();
    }
}

```

Output

1. Bean Object Created
2. Initialization Method Called
3. Bean Ready For Use
4. Destruction Method Called

Bean Life Cycle Flow

- ♦ Spring Container Starts
- ♦ Bean Object Created (Constructor Called)
- ♦ Initialization Method Executed
- ♦ Bean Ready For Use
- ♦ Container Closed
- ♦ Destruction Method Executed

Interview Point

- ♦ Bean Life Cycle refers to the sequence of steps a bean undergoes from creation to destruction inside the Spring IoC Container.

- ♦ In this example:

👉 Constructor → Bean Creation

👉 initMethod → Initialization

👉 Business Method → Bean Usage

👉 destroyMethod → Bean Destruction